Question 1 (25 marks)

A smart warehouse robot executes a route-planning program consisting of 4.8×10^9 instructions. The processor has an average CPI of 2.5 and operates at a clock rate of 3 GHz. During system evaluation, engineers compare the processor against a newer model that completes the same workload in 2.5 seconds.

(a) Analyse the processor performance by determining:

(i) The execution time of the original processor.

(ii) The speedup achieved by the newer processor.

(15 Marks)

(b) The warehouse manager claims that doubling the clock rate will always double system performance.

Evaluate *this claim by comparing the roles of instruction count, CPI, and clock rate in determining execution time. Support your answer using the CPU performance equation.*

0 Marks)

Question 2 (25 marks)

A drone navigation controller repeatedly evaluates the expression

$$f = (g + h) - (i + j)$$

where:

Variable	Register
g	$s0
h	$s1
i	$s2
j	$s3
f	$s4

(a) Break down the expression into valid MIPS instructions and **determine** the minimum number of instructions required to complete the operation.

(15 Marks)

(b) The processor executes the instruction

add $t0,$s0,$s1

Given:

- *opcode = 000000*
- *rs = 10000*
- *rt = 10001*
- *rd = 01000*
- *shamt = 00000*
- *funct = 100000*

Determine the complete 32-bit machine code representation and **distinguish between** the roles of the opcode and function fields during instruction execution (Hint: use appendix information for this question).

(10 Marks)

Question 3: (25 Marks)

A smart manufacturing controller accesses data through a cache memory with an access time of 2 ns and a main memory with an access time of 80 ns. During testing, the cache achieves a hit rate of 95%.

(a) Analyse the memory hierarchy by determining:

(i) The cache miss rate.

(ii) The Average Memory Access Time (AMAT).

(15 Marks)

(b) A second cache design increases the hit rate to 98% but increases cache access time to 4 ns. **Compare** the two cache designs by determining the AMAT for each design and **evaluate** which design should be selected for deployment.

(10 Marks)

Question 4: (25 Marks)

An autonomous exploration rover executes a mission-control application containing the following instruction mix:

Instruction Type	Instruction Count	CPI
Arithmetic	400 million	1
Memory Access	350 million	5
Branch	250 million	3

The processor operates at a clock rate of 2.5 GHz.

To improve performance, engineers propose a hardware enhancement that reduces the CPI of memory-access instructions from 5 to 2.

(a) **Analyse** the original and enhanced processors by determining:

- The total CPU cycles required before enhancement.
- The execution time before enhancement.
- The execution time after enhancement.
- The overall speedup achieved.

(15 Marks)

(b) Further analysis reveals that memory-access instructions account for 70% of the original execution time. A second proposal introduces a specialized memory accelerator that makes memory-access operations 8 times faster.

Using Amdahl's Law, **determine** the maximum achievable system speedup and **evaluate** which improvement strategy should be prioritized: CPI reduction or memory acceleration. Support your recommendation using quantitative evidence obtained from your analysis.

(10 Marks)

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